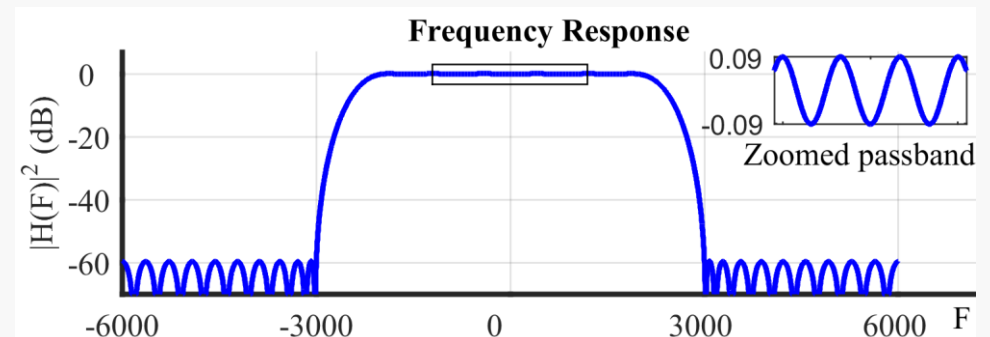
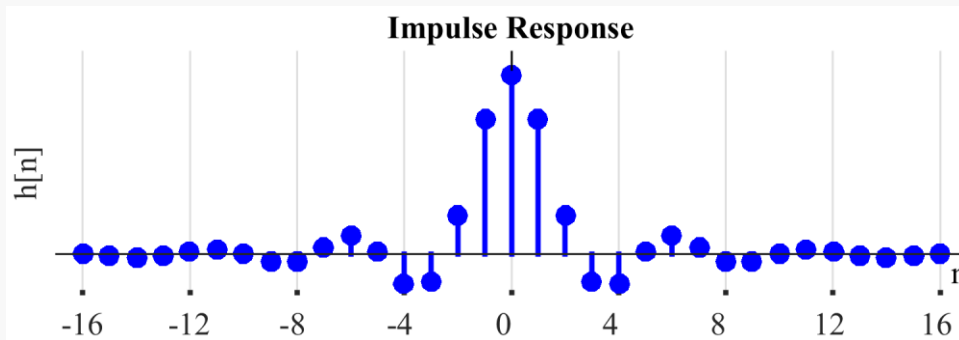


Introduction to Systems

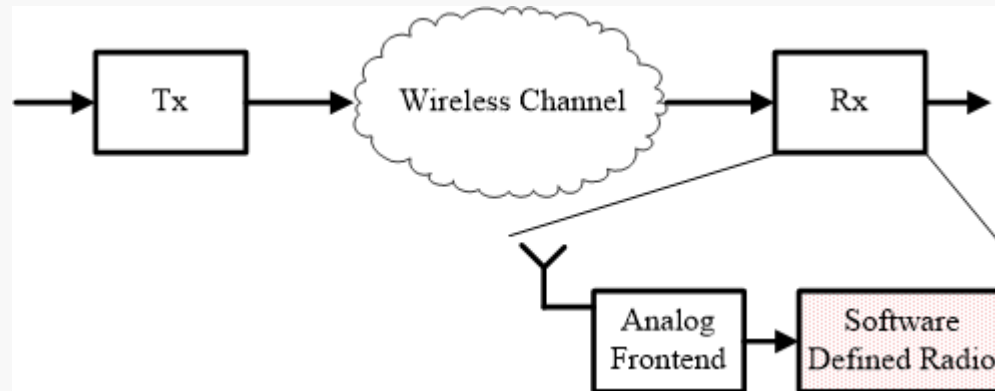
Qasim Chaudhari



System

A device/algorithm that performs some prescribed operations on an input signal to generate an output signal

Tx, wireless channel and Rx are all systems



Linear System

- Scaling and addition



- Output of a linear system contains the same set of frequencies, remember

$$(a + b)^2 = a^2 + b^2 + 2ab$$

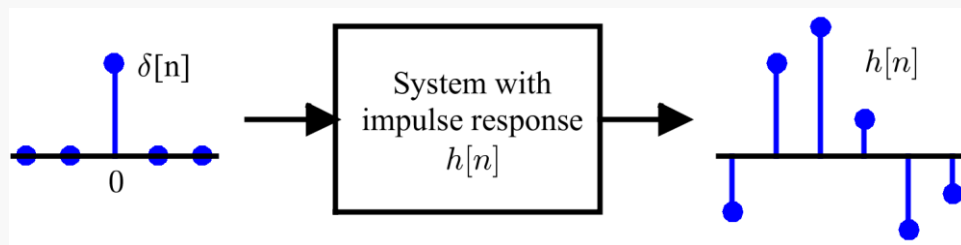
Time Invariance

If the input is shifted by a certain time, the output is also shifted (but otherwise the same)

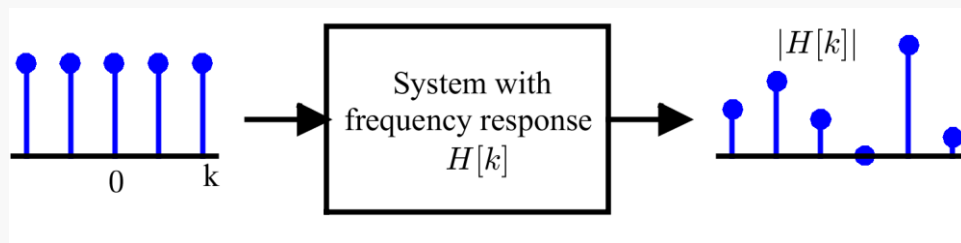
$$\text{input } s[n - n_0] \longrightarrow r[n - n_0] \text{ output}$$

System Characterization

Impulse response $h[n]$: output response of a system to a unit impulse $\delta[n]$ as an input

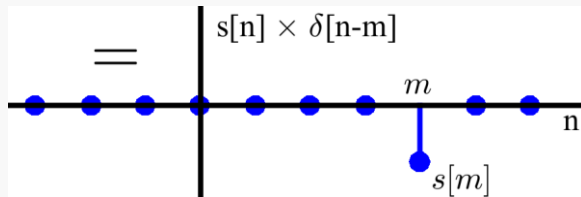
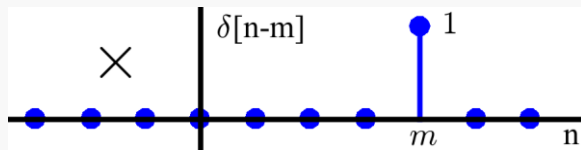
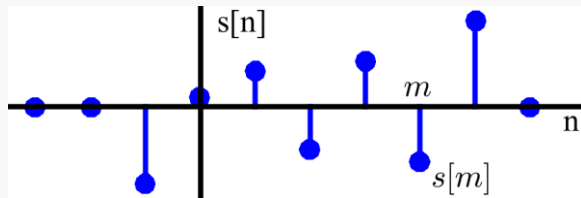


Frequency response $H[k]$: The DFT of the system impulse response



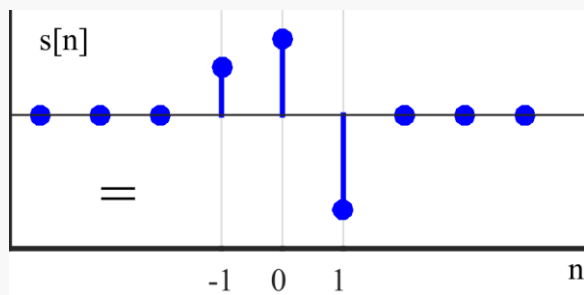
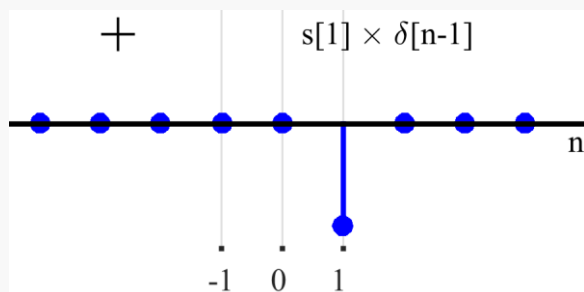
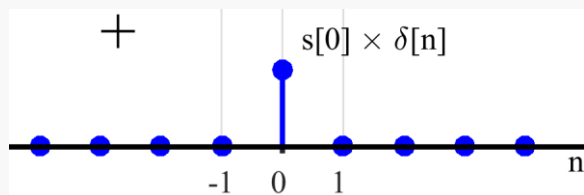
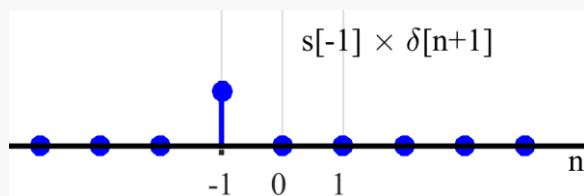
What is the DFT of a unit impulse?

Let Us Decompose a Signal



$$s[n]\delta[n-m] = s[m]\delta[n-m]$$

Let Us Decompose a Signal



$$\begin{aligned}
 s[n] &= \cdots + s[-2]\delta[n+2] + s[-1]\delta[n+1] + \\
 &\quad s[0]\delta[n] + s[1]\delta[n-1] + s[2]\delta[n-2] + \cdots \\
 &= \sum_{m=-\infty}^{\infty} s[m]\delta[n-m]
 \end{aligned}$$

Convolution

input	→	output	
$\delta[n]$	→	$h[n]$	
$\delta[n - m]$	→	$h[n - m]$	(time-invariance)
$\alpha_1 \cdot \delta[n - m]$	→	$\alpha_1 \cdot h[n - m]$	(linearity)
$\alpha_2 \cdot \delta[n - m']$	→	$\alpha_2 \cdot h[n - m']$	(linearity)
$\alpha_1 \cdot \delta[n - m] + \alpha_2 \cdot \delta[n - m']$	→	$\alpha_1 \cdot h[n - m] + \alpha_2 \cdot h[n - m']$	(LTI)

This leads to input output sequence as

+		→		+
+	$s[-2] \cdot \delta[n + 2]$	→	$s[-2] \cdot h[n + 2]$	+
+	$s[-1] \cdot \delta[n + 1]$	→	$s[-1] \cdot h[n + 1]$	+
+	$s[0] \cdot \delta[n]$	→	$s[0] \cdot h[n]$	+
+	$s[1] \cdot \delta[n - 1]$	→	$s[1] \cdot h[n - 1]$	+
+	$s[2] \cdot \delta[n - 2]$	→	$s[2] \cdot h[n - 2]$	+
+		→		+
=	$s[n]$	→	$\sum_{m=-\infty}^{\infty} s[m]h[n - m]$	=

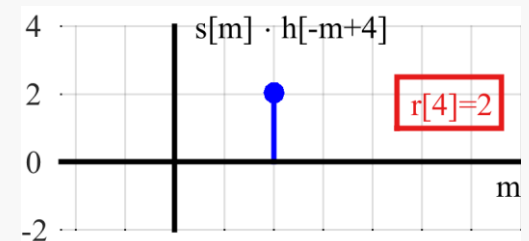
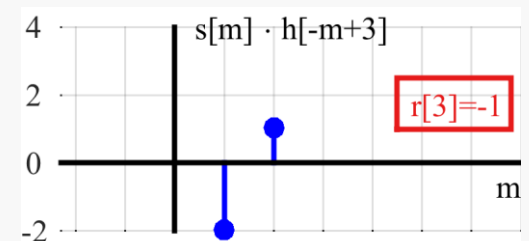
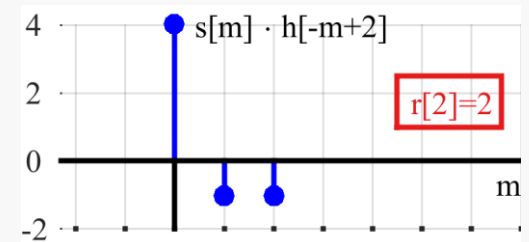
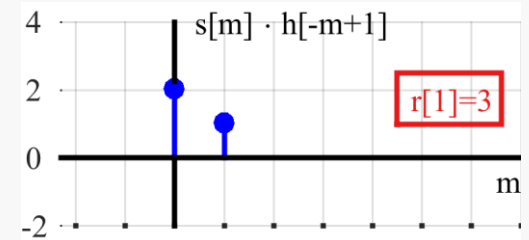
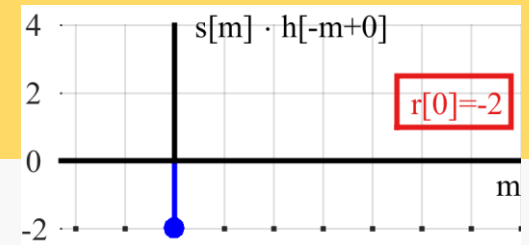
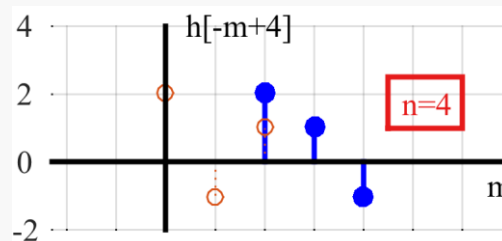
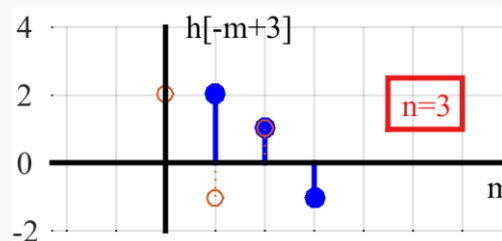
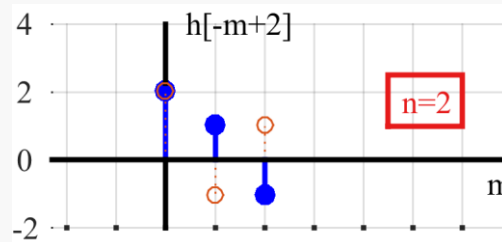
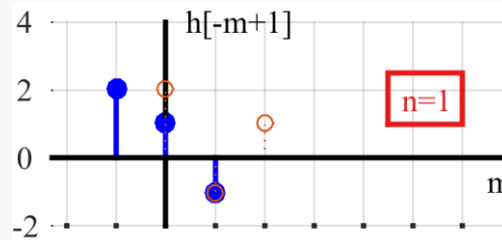
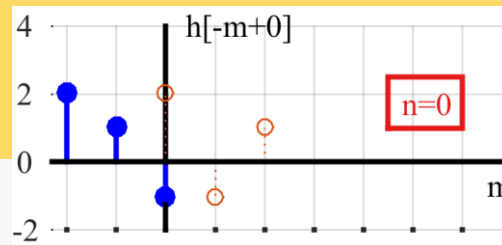
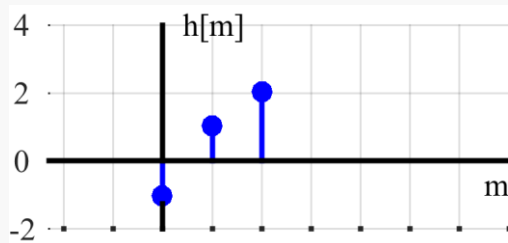
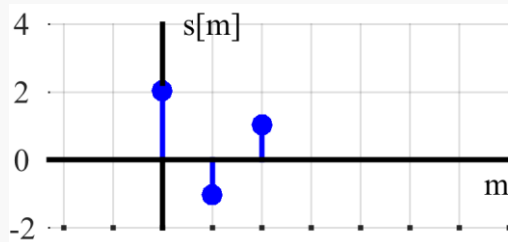
Convolution

- For two signals $s[n]$ and $h[n]$, convolution is defined as

$$r[n] = \sum_{m=-\infty}^{+\infty} s[m]h[n-m]$$

1. Select the time shift n
2. Compute the sample-by-sample product $s[m]h[n-m]$ and add for all m
3. Repeat for all n

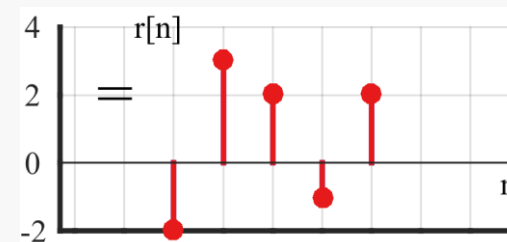
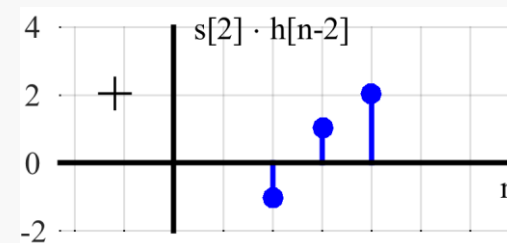
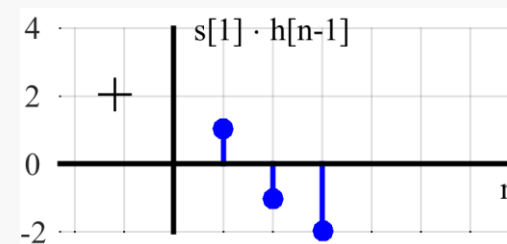
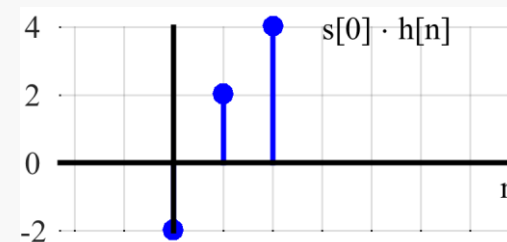
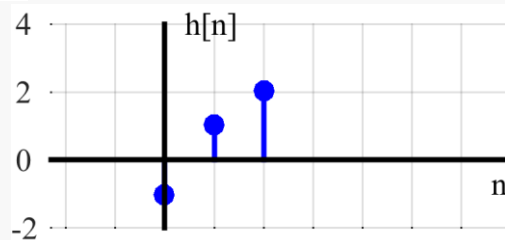
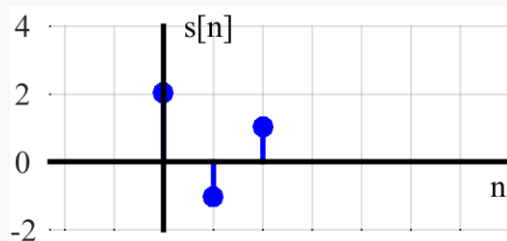
Example



Intuitive Method

Straight from the definition

$$r[n] = \sum_{m=-\infty}^{\infty} s[m]h[n-m]$$



Fundamental Relation of DSP

Convolution in one domain is equivalent to multiplication in the other domain

$$s[n] \circledast h[n] \xrightarrow{\text{F}} S[k] \cdot H[k]$$

The importance of this relation cannot be overemphasized!

Role of Complex Sinusoids

Let us excite an LTI system with a real impulse response $h[n]$ by an input complex sinusoid $V[n]$

$$V_I[n] = \cos 2\pi \frac{k}{N} n$$

$$V_Q[n] = \sin 2\pi \frac{k}{N} n$$

Then, the system output is $r[n] = H[k] \cdot V[n]$ which can be written as

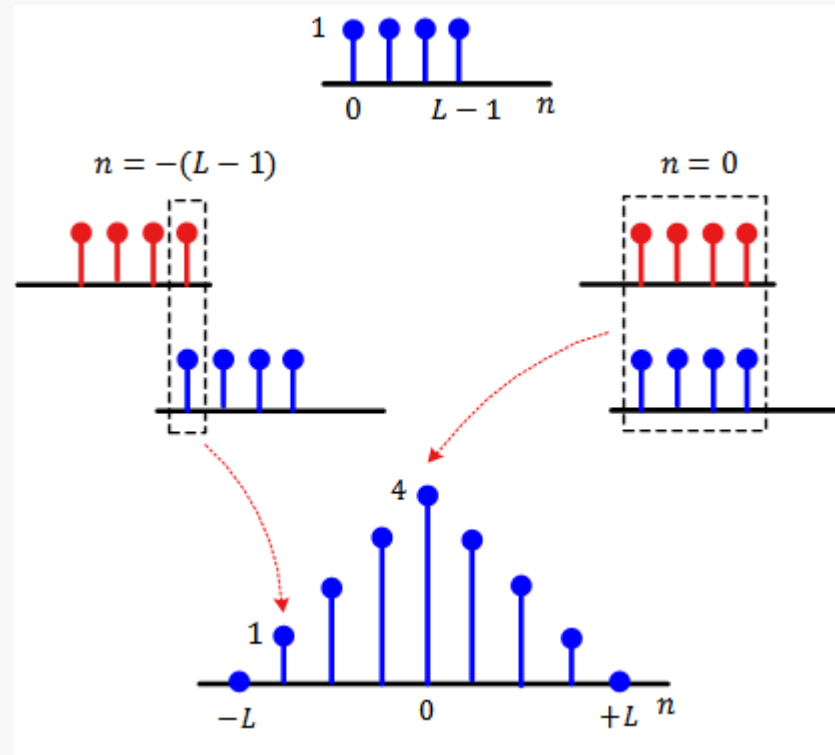
$$r_I[n] = |H[k]| \cdot \cos \left(2\pi \frac{k}{N} n + \angle H[k] \right)$$

$$r_Q[n] = |H[k]| \cdot \sin \left(2\pi \frac{k}{N} n + \angle H[k] \right)$$

Autocorrelation

When a signal is correlated with itself, it is called **auto-correlation**

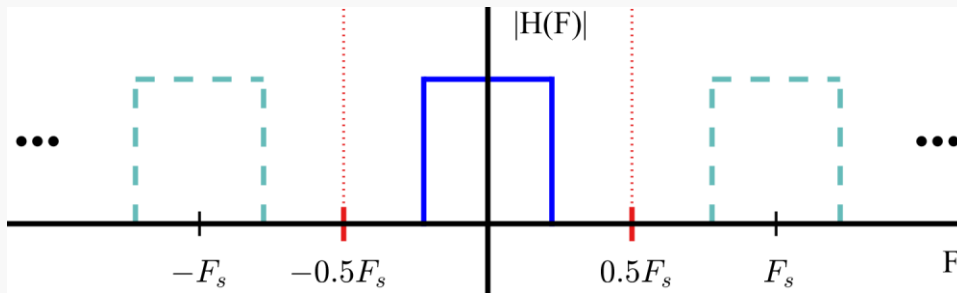
Example: a rectangular pulse



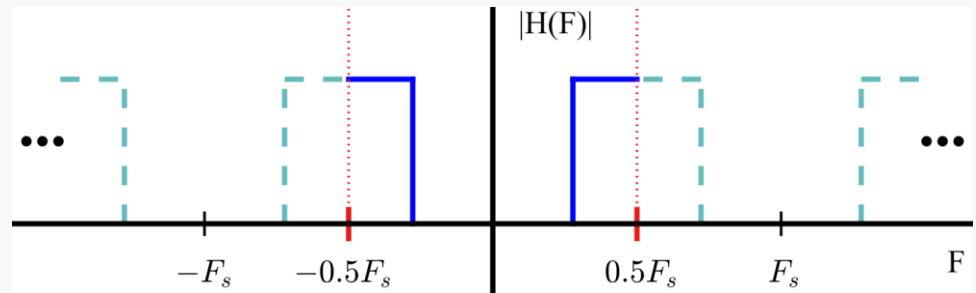
FIR Filters

A filter modifies the frequency contents of an input signal

Lowpass filter



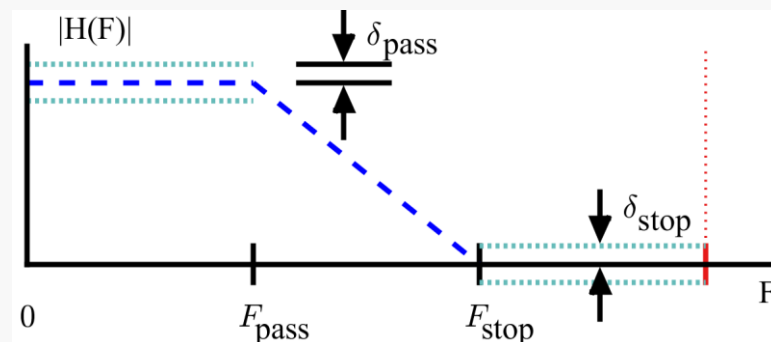
Highpass filter



Finite impulse response (FIR) filters

FIR Filter Design

- Sample rate F_S
- Passband frequency F_{pass}
- Stopband frequency F_{stop}
- Maximum passband ripple δ_{pass}
- Maximum stopband ripple δ_{stop}



Matlab command

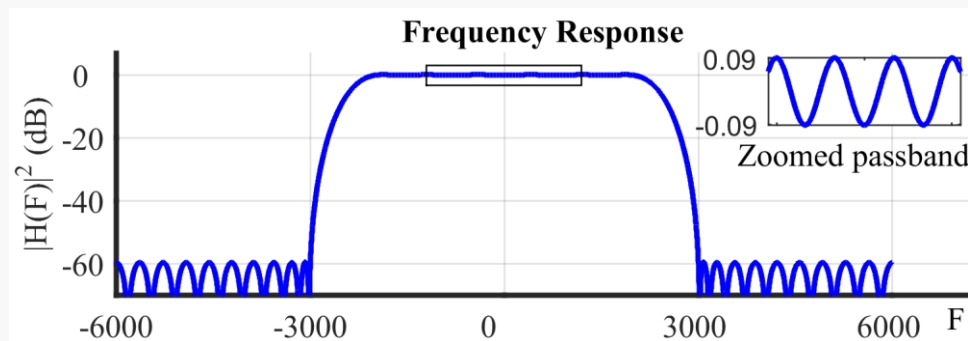
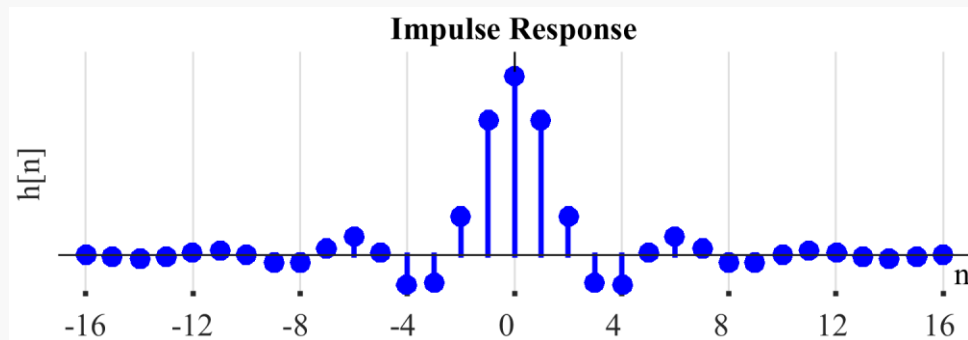
$$h = \text{firpm}(N - 1, [0 \quad F_{\text{pass}} \quad F_{\text{stop}} \quad 0.5F_S]/(0.5F_S), [1 \quad 1 \quad 0 \quad 0])$$

FIR Filter Design

- Sample rate $F_S = 12$ kHz
- Passband frequency $F_{\text{pass}} = 2$ kHz
- Stopband frequency $F_{\text{stop}} = 3$ kHz
- Passband ripple $\delta_{\text{pass,dB}} = 0.1$ dB
- Stopband ripple $\delta_{\text{stop,dB}} = -60$ dB

Matlab command

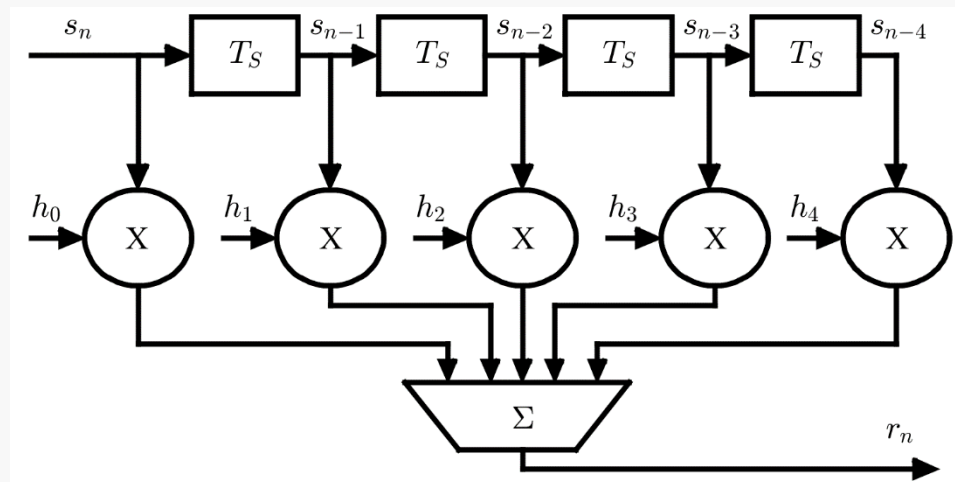
$$h = \text{firpm}(28, [0 \ 2000 \ 3000 \ 6000]/6000, [1 \ 1 \ 0 \ 0])$$



FIR Filter Structure

$$r[n] = \sum_{m=0}^{L_{\text{FIR}}-1} h[m]s[n-m]$$

$$= h[0]s[n] + h[1]s[n-1] + h[2]s[n-2] + h[3]s[n-3] + h[4]s[n-4]$$

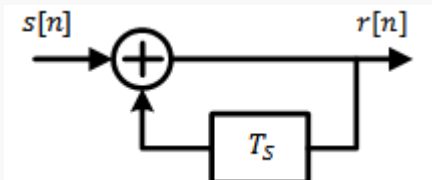
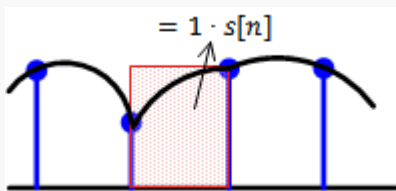


Discrete-Time Integrator

Forward difference

- Computes its output after the arrival of the current sample

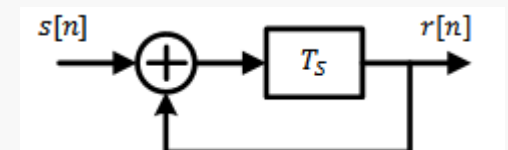
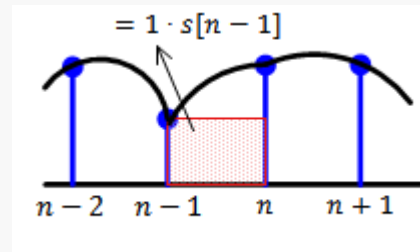
$$r[n] = r[n - 1] + s[n]$$



Backward difference

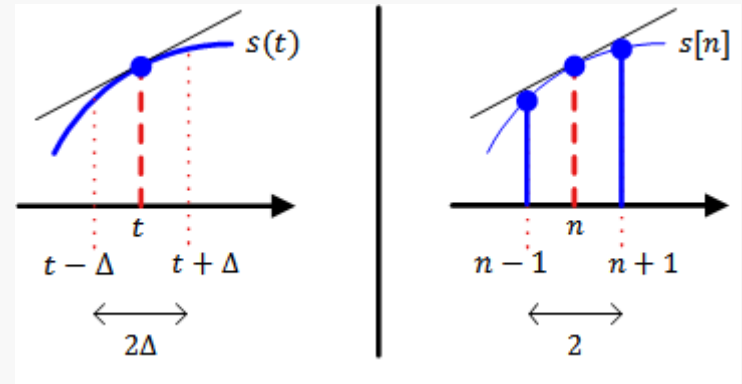
- Computes its output after the last sample

$$r[n] = r[n - 1] + s[n - 1]$$



Differentiator

- Derivative is defined as the slope of the line tangent to the curve at a point

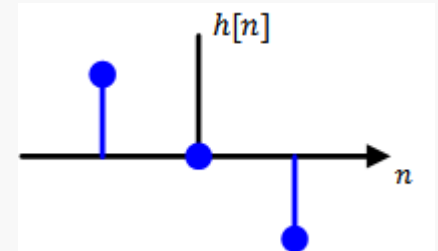


- In discrete-time, we approximate it as

$$\dot{s}[n] = \frac{s[n + 1] - s[n - 1]}{2}$$

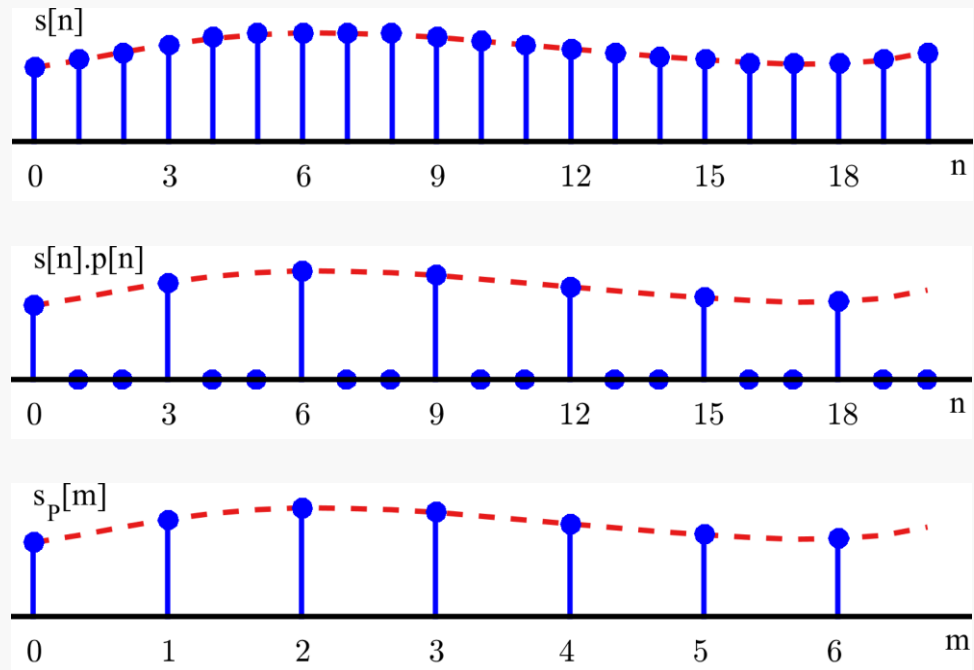
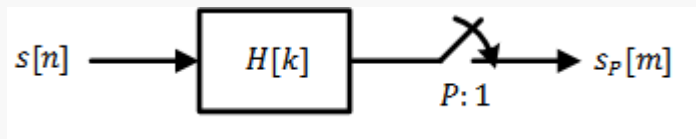
- Discrete-time filter coefficients

$$h[n] = \frac{1}{2} \{+1, 0, -1\}$$



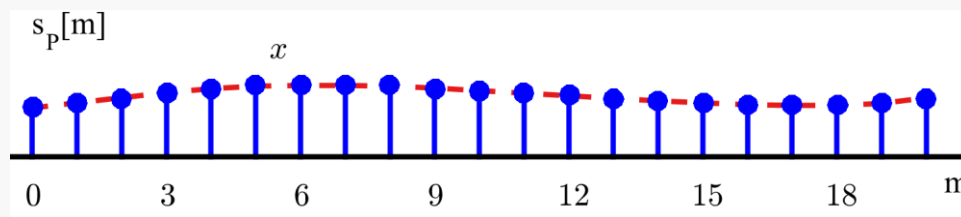
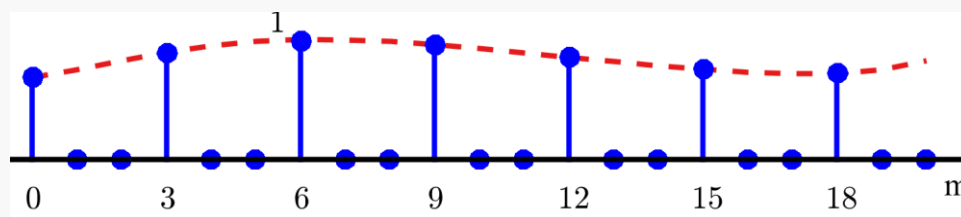
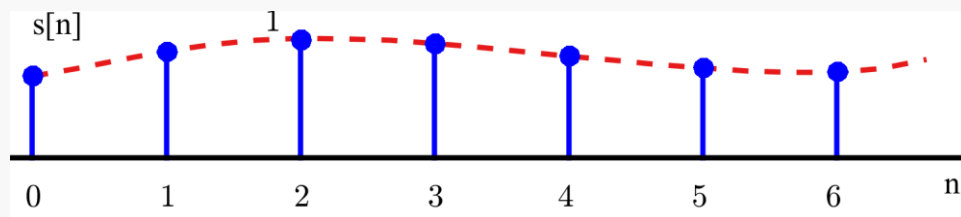
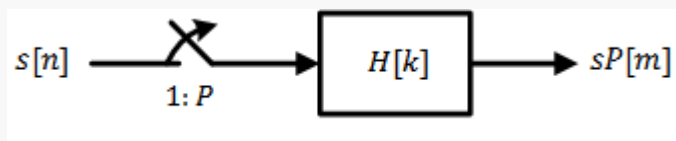
Downsampling

- Reducing the sample rate by an integer factor
- Retain every P^{th} sample and discard the remaining samples
- The new slower sample rate is $1/P$ of the original faster sample rate



Upsampling

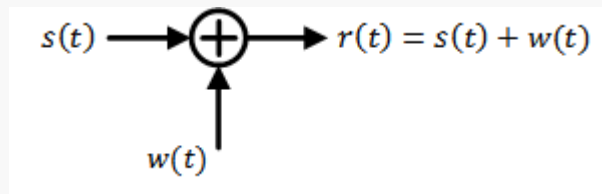
- Increasing the sample rate by an integer factor
- Insert $P - 1$ zeros between the original input samples and interpolate
- The new faster sample rate is P times the original slower sample rate



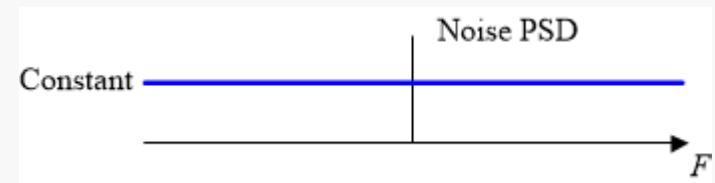
Additive White Gaussian Noise (AWGN)

Addition of random signals arising from the vibration of atoms in the receiver electronics

Additive



White



Gaussian

